

PUBLICATIONS

International Conferences:

- Gabriele Lobbia, Wojciech Różowski, Ralph Sarkis, and Fabio Zanasi. *Quantitative Monoidal Algebra: Axiomatising Distance with String Diagrams*. In: [MFCS '25](#).
- Ralph Sarkis and Fabio Zanasi. *String Diagrams for Graded Monoidal Theories with an Application to Imprecise Probability*. In: [CALCO '25](#).
- Matteo Mio, Ralph Sarkis, and Valeria Vignudelli. *Beyond Nonexpansive Operations in Quantitative Algebraic Reasoning*. In: [LICS '22](#).
- Daniela Petrişan and Ralph Sarkis. *Semialgebras and Weak Distributive Laws*. In: [MFPS '21](#).
- Matteo Mio, Ralph Sarkis, and Valeria Vignudelli. *Combining Nondeterminism, Probability and Termination: Equational and Metric Reasoning*. In: [LICS '21](#).

Journals:

- Matteo Mio, Ralph Sarkis, and Valeria Vignudelli. *Universal Quantitative Algebra for Fuzzy Relations and Generalised Metric Spaces*. In: [LMCS](#).

Preprints:

- Ralph Sarkis and Fabio Zanasi. *Graded String Diagrams for Imprecise Probability and Causal Intervention*. Submitted to LMCS after invitation from CALCO '25 committee.
- Ralph Sarkis and Fabio Zanasi. *Complete Diagrammatic Axiomatisations of Relative Entropy*. Submitted to MFPS '26.

EDUCATION

Theoretical Computer Science (PhD) ENS de Lyon Thesis title: <i>Lifting Algebraic Reasoning to Generalized Metric Spaces</i> . Available here .	September 2024 Lyon, France
Theoretical Computer Science (Master) ENS de Lyon Funded by two 10000€ scholarships from Labex MiLyon.	June 2021 Lyon, France
Joint Honours in Math and Computer Science (BSc) McGill University	April 2019 Montreal, QC

WORK EXPERIENCE

Academic Positions:

- Postdoctoral research fellow at PPLV (University College London). October 2024 – Ongoing
 - Work on diagrammatic reasoning for probabilistic programs with Fabio Zanasi.
- Research internship at IRIF (Université Paris Cité). February – June 2021
 - Work on weak distributive laws supervised by Daniela Petrişan.
- Research internship at LIP (ENS de Lyon). April – July 2020
 - Work on algebraic presentations of monads supervised by Matteo Mio and Valeria Vignudelli.

Teaching:

- Teaching assistant at University College London. 2025–2026
 - Tutorials for *(Co)algebraic Structures in Computer Science* and *Discrete Mathematics for Computer Science* (total of 25 hours in class).
 - Grading reports and presentations for a directed reading course.
- Teaching assistant at ENS de Lyon. 2021 – 2024
 - Tutorials for *Semantics and Verification* and *Proofs and Programs* (total of 100 hours in class).
- Teacher and organizer for a category theory student seminar at McGill and ENS de Lyon. 2019 – 2023
 - Lectures, [lecture notes](#), and homework for a semester-long introduction to category theory.
 - Supervision of two cohorts of student lecturers at ENS de Lyon.
- Teaching assistant at McGill University. September – December 2018
 - Office hours and grading for *Theory of Computation*.

Outreach:

- Mini-course at [North Bridge House Senior Hampstead](#) July 2025
 - Organizing and teaching a 10 hours course (over 2 days) on ZX calculus.

- Volunteer animator at **MMI**. 2023 – 2024
– Workshops and exhibition *Dans ma cuisine* (on math and CS) for students ages 13 to 18.
- Volunteer internship supervisor for **UPBS**. April 2022, 2023, and 2024
– Week-long initiation to research on string diagrams for students ages 16 to 17.
- Instructor and teaching assistant at **DMA**. July 2015 and 2017
– Programming and electronics summer camp classes for students ages 8 to 17.

INVITED (★) AND CONTRIBUTED TALKS

- ★ *First-Order Logic and Trees of Graphs*, **Aminfest**. December 2025
- *Towards Quantitative Diagrammatic Reasoning*, **GT Scalp**. October 2025
- *Tutorial: String Diagrams for CS*, **GT Scalp**. October 2025
- *Towards Quantitative Diagrammatic Reasoning*, **MFCS conference**. August 2025
- *Imprecise Probability with Graded String Diagrams*, **CALCO conference**. June 2025
- ★ *Graded String Diagrams*, **OASIS seminar**. March 2025
- ★ *Quantitative Monoidal Algebra*, Bellairs workshop *Quantitative Reasoning*. March 2025
- *Lifting Algebraic Reasoning to Generalized Metric Spaces*, online **CT Zulip** seminar. August 2024
- *Quantitative Algebraic Semantics*, **InfoMaths** doctoral school days. June 2024
- ★ *Universal Quantitative Algebra*, **LIMD team seminar**. October 2023
- ★ *Universal Quantitative Algebra*, **LoVE team seminar**. September 2023
- ★ *Lifting Algebraic Presentations to Met*, Bellairs workshop *Quantitative Logic and Reasoning*. May 2023
- *I show you my favorite diagrams (the 3rd is ULTRA rare!)*, **McGill grad seminar**. October 2022
- *Tape Diagrams*, **ACT adjoint school**. July 2022
- *Je vous montre mes plus beaux diagrammes (le 3e va vous surprendre !)*, **CGS general public talk**. May 2022
- *Quotienting a Monad via Projective Algebras*, **RAMiCS conference**. November 2021
- *Semialgebras and Weak Distributive Laws*, **MFPS conference**. September 2021
- *Modelling Nondeterminism, Probability, and Termination*, **LiCS conference**. July 2021
- *Modelling Nondeterminism, Probability, and Termination*, **CaCS workshop**. June 2021

ACADEMIC SERVICES

- I was a peer reviewer for several prestigious conferences (FSCD 2023, FoSSaCS 2023, LiCS 2024, ICALP 2025, MFPS 2025, MFCS 2025, CSL 2026, ESOP 2026, and FoSSaCS 2026) as well as the journal *Logical Methods in Computer Science* (**LMCS**) and the *Polynesian Journal of Mathematics*.
- I am a TA for the **Adjoint School 2026** project led by Ana Sokolova on *Variants of (Coalgebraic) Simulations*.